Applied Statistics for Engineers and Scientists II

Lab 02 — Practice Problems

Polynomial Regression, Transformations, Ridge, and Lasso

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Exercise 1. Transformation to linearize an exponential relationship

A laboratory records the number of surviving bacteria after radiation exposure:

time t	0	1	2	3	4	5
count N_t	100	80	65	51	41	33

Assume an exponential mean law

$$N_t = N_0 e^{\beta t}, \qquad N_0 > 0.$$

(a) Linearize the model by applying an appropriate transformation. Write down the resulting linear model and identify the new intercept in terms of the original parameters.

(b) Fit the transformed regression in R using ordinary least squares. Report the estimated slope $\widehat{\beta}$ and the estimated initial count $e^{\widehat{\alpha}}$.

(c) Interpret the slope on the original (count) scale. What is the estimated multiplicative change in expected count for one additional time unit? What is the estimated proportional decrease per time unit?

(d) Check the residuals on the log scale. Explain why residual diagnostics should be performed on the transformed scale rather than the original count scale.

Exercise 2. Ridge shrinkage

Suppose a centered standardized design satisfies

$$X^\top X = \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 9 \\ 9 \end{pmatrix}.$$

(a) Compute the ordinary least-squares estimate $\widehat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y$.

(b) Compute the ridge estimate for $\lambda = 2$:

$$\widehat{\beta}_\lambda = (X^\top X + \lambda I_2)^{-1} X^\top y.$$

(c) Compute the eigenvalues of $X^\top X$ and of $X^\top X + 2I_2$.

(d) Compute the condition number $\kappa_2(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ for both matrices. Explain why adding $2I_2$ improves numerical stability.

Exercise 3. Lasso soft-thresholding

Assume
$$X^\top X = I_4, \qquad X^\top y = (3,\ 0.7,\ -1.2,\ -4)^\top.$$

(a) Using the soft-thresholding formula
$$\widehat{b}_j = \operatorname{sign}(z_j)\big(|z_j| - \lambda\big)_+, \qquad z = X^\top y,$$
compute the lasso solution $\widehat{b}_\lambda$ for $\lambda = 1$. Which variables are selected (i.e. have nonzero coefficients)?

(b) Repeat the computation for $\lambda = 2$. Which variables are selected now?

(c) Explain why increasing λ makes the selected model smaller (i.e. fewer variables are retained).